

# Moons & Ring Systems of the Solar System

## Major Moons of the Solar System

---

Several planets have notable moons. The diagram below compares the largest moons.

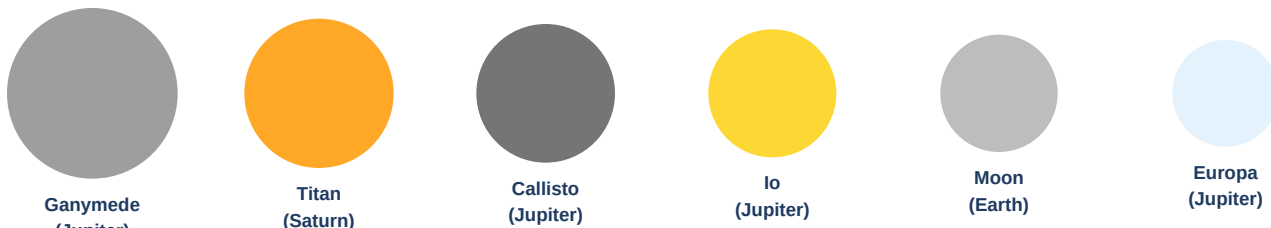


Figure 3: Relative sizes of the six largest moons in the Solar System.

## Moon Facts

---

- Ganymede (Jupiter): Diameter 5,268 km. Larger than the planet Mercury. Has its own magnetic field.
- Titan (Saturn): Only moon with a dense atmosphere. Surface pressure 1.5x Earth's. Has methane rain.
- Europa (Jupiter): Surface covered in ice. Scientists believe a liquid water ocean exists beneath.
- Io (Jupiter): Over 400 active volcanoes. Heated by tidal forces from Jupiter's gravity.
- The Moon (Earth): Stabilizes Earth's axial tilt. Distance: 384,400 km. Orbital period: 27.3 days.
- Triton (Neptune): Orbits Neptune in reverse. Likely a captured Kuiper Belt object.

# Ring Systems & The Greenhouse Effect

## Planetary Ring Systems Diagram

Four planets in our Solar System have ring systems: Saturn, Jupiter, Uranus, and Neptune.



Figure 4: Ring systems of the four ringed planets (not to scale).

## Saturn's Ring System — Key Facts

- Saturn has the most prominent ring system, visible even through a small telescope.
- The rings extend up to 282,000 km from Saturn but are only about 10 meters thick on average.
- Rings are made of billions of chunks of ice and rock ranging from tiny grains to house-sized boulders.
- Saturn has 7 main ring groups labeled A through G.
- The rings are thought to be remnants of a destroyed moon or comet captured by Saturn's gravity.

## The Greenhouse Effect on Venus

Venus has a runaway greenhouse effect making it the hottest planet despite not being closest to Sun.

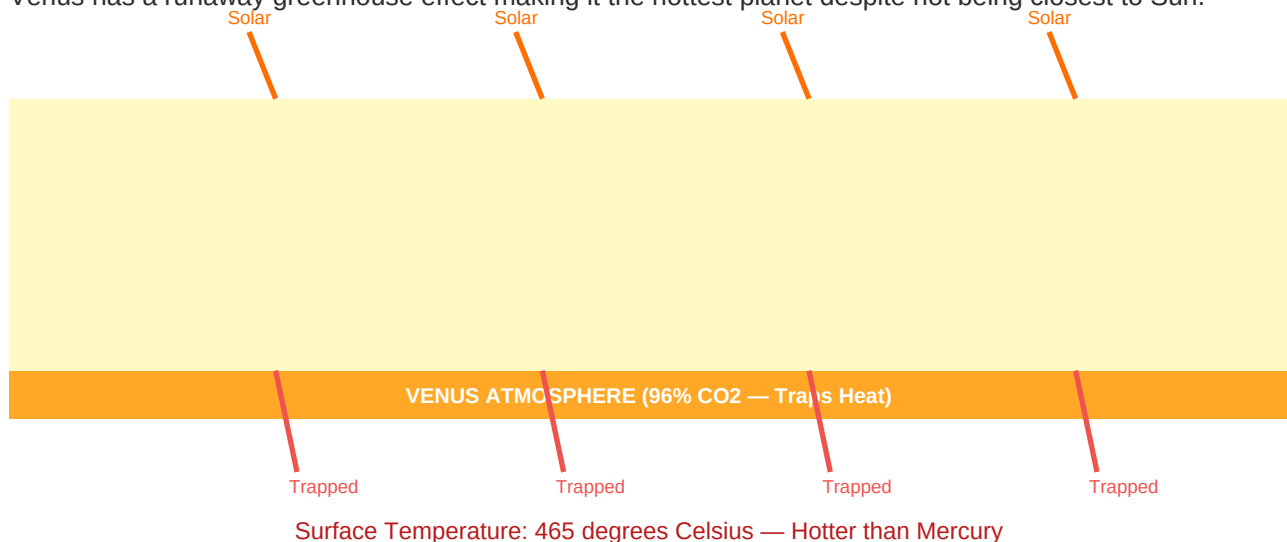


Figure 5: Simplified greenhouse effect diagram for Venus.